Workmanship acceptance specification: PCB1 and PCB4 assemblies

Demonstration document WS-PCB-014, revision A, 10 October 2026. Written for the Spec to Inspector entry to the ASUS UGen AI League hackathon, in the style of a customer's workmanship specification. It is not a real customer's document: its parts are the boards photographed in the public VisA dataset, its limits are illustrative, and it neither quotes nor paraphrases any published standard.

1 Scope

This specification states the visual acceptance criteria for two assembled circuit boards supplied to the customer: part PCB1, an ultrasonic ranging module, and part PCB4, a single-cell charger module with a micro USB receptacle. It applies at final visual inspection of the component side of each board, from one image per board taken from directly above.

Each defect found on a board is judged on its own against the one clause that names its defect class and the zone it lies in. A board is conforming when no clause rejects any of its defects; one rejected defect rejects the board. A defect that fits no clause of its part, or that lies outside every zone, is referred to the customer's quality engineer for review and is neither accepted nor rejected by the supplier.

2 Definitions

- **Defect class**: the name of a kind of defect, written in lower case exactly as inspection records carry it. The classes of a part are those named in its clauses.
- **Size**: the longest extent of a defect, which is the long side of the smallest rectangle, at any angle, that encloses the defect's visible area. Every limit given in millimetres in this document is a limit on size.
- **Zone**: a named rectangle of the inspection image, declared in section 3.2. A defect belongs to the zone its centre lies in. Where zones overlap, or a defect lies across more than one zone, the most critical of those zones applies.
- **Any**: written in place of a zone, it means the clause applies wherever on the part the defect lies.
- **Limit**: the bound on size beyond which a defect is nonconforming. "1.0 mm max" means a size up to and including 1.0 mm is acceptable and a larger one is not. "> 1.0 mm" means the defect is nonconforming when its size is greater than 1.0 mm. The two spellings state the same rule. "Not permitted" means the defect is nonconforming at any size.
- **Disposition**: what is done with a nonconforming defect. "Reject" means the board is rejected and is not shipped.

3 Calibration and zones

3.1 Calibration

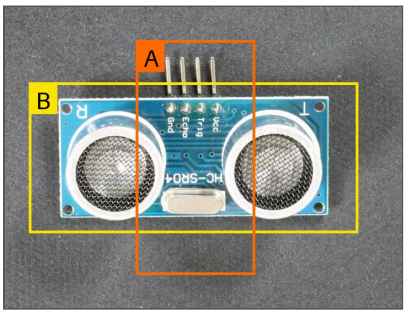
Each part is imaged from directly above at a fixed working distance, with the whole board in the frame and a margin of fixture around it. The table declares the nominal width of each board and the width and height of the scene that the full image covers. A horizontal length in the image is converted to millimetres by the imaged width divided by the image's width in pixels, and a vertical length by the imaged height divided by the image's height in pixels.

Part	Board width (mm)	Imaged width (mm)	Imaged height (mm)
PCB1	45.0	66.9	51.0
PCB4	25.0	46.7	38.0

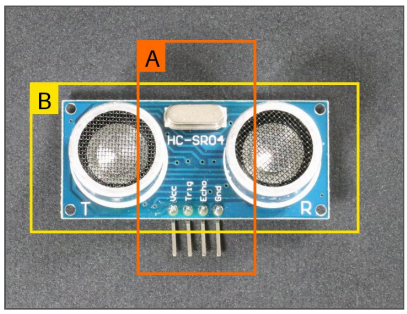
3.2 Zones

Zones are rectangles in normalised image coordinates. The value x runs from 0 at the left edge of the image to 1 at its right edge, and y runs from 0 at the top edge to 1 at the bottom edge. The corner (x0, y0) is a zone's top left and (x1, y1) its bottom right. The zones of each part are listed with the most critical first. Part PCB1 is presented to the camera either way round, with its header pins towards the top or towards the bottom of the image, so its zones are drawn to hold in both presentations.

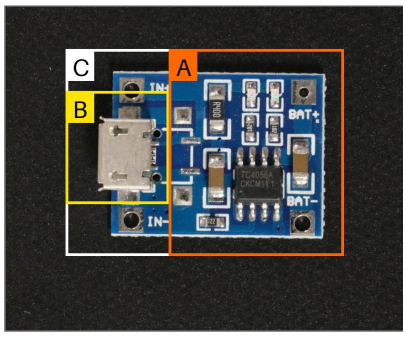
Part	Zone	Covers	x0	y0	x1	y1
PCB1	A	Header pins, their solder joints and the crystal	0.33	0.11	0.63	0.89
PCB1	B	Board face and transducers	0.06	0.25	0.89	0.75
PCB4	A	Components and their solder joints	0.41	0.13	0.85	0.77
PCB4	B	USB receptacle	0.15	0.26	0.41	0.61
PCB4	C	Input pads and board margin	0.15	0.13	0.85	0.77



Zones of PCB1, header pins towards the top (VisA pcb1, normal 0272).



Zones of PCB1, header pins towards the bottom (VisA pcb1, normal 0630).



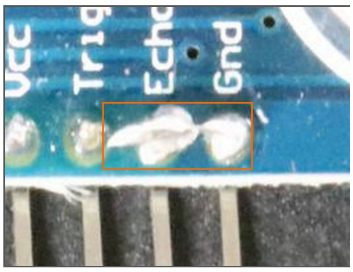
Zones of PCB4 (VisA pcb4, normal 0537).

4 Acceptance criteria for part PCB1

4.1 Stray or bridging solder

Solder that has flowed or splashed to where the design puts none is nonconforming at any size and in any zone. This covers a lump or bridge of solder joining two header pins or two pads, and solder lying on a component body or on the bare board.

Clause	Defect class	Zone	Limit	Disposition
4.1	melt	Any	Not permitted	Reject

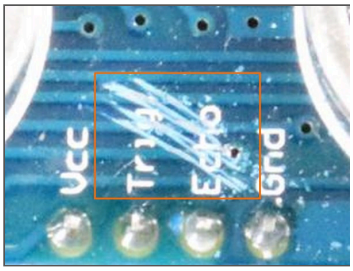


Solder bridging two header pads (VisA pcb1, anomaly 024).

4.2 Scratch

A scratch is a line cut or scored into the surface of the board. It is nonconforming when its size exceeds the limit of the zone it lies in. Several scratches that touch or cross are measured together as one defect.

Clause	Defect class	Zone	Limit	Disposition
4.2.1	scratch	A	1.0 mm max	Reject
4.2.2	scratch	B	2.0 mm max	Reject

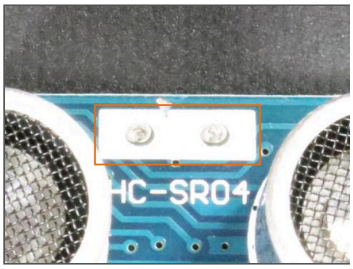


Scratches beside the header (VisA pcb1, anomaly 090).

4.3 Missing component

Every component of the assembly shall be present: both transducers with their mesh covers, the crystal and the four header pins. An absent component, or a transducer without its cover, is nonconforming in any zone.

Clause	Defect class	Zone	Limit	Disposition
4.3	missing	Any	Not permitted	Reject

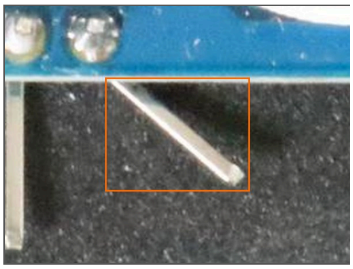


The crystal absent from its pads (VisA pcb1, anomaly 042).

4.4 Bent header pin

The four header pins shall be straight, parallel to one another and square to the board edge. A pin bent out of line with its neighbours, or touching another pin, is nonconforming at any angle of bend.

Clause	Defect class	Zone	Limit	Disposition
4.4	bent	Any	Not permitted	Reject



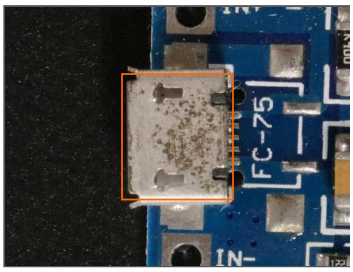
A header pin bent across its neighbour (VisA pcb1, anomaly 003).

5 Acceptance criteria for part PCB4

5.1 Foreign material

Foreign material is anything on the part that does not belong to it and is not a component: dirt, residue, corrosion spots or tarnish. It is nonconforming when its size exceeds the limit of the zone it lies in. Spots that touch are measured together as one defect.

Clause	Defect class	Zone	Limit	Disposition
5.1.1	dirt	A	> 0.5 mm	Reject
5.1.2	dirt	B	> 1.0 mm	Reject
5.1.3	dirt	C	> 2.0 mm	Reject

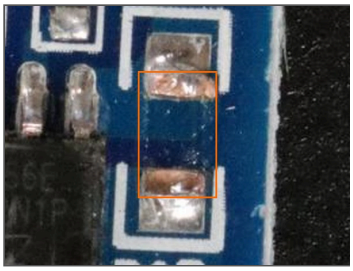


Corrosion spots across the receptacle shell (VisA pcb4, anomaly 080).

5.2 Missing component

Every component of the assembly shall be present on its pads: the receptacle, the charger integrated circuit, and each resistor, capacitor and indicator. An absent component is nonconforming in any zone.

Clause	Defect class	Zone	Limit	Disposition
5.2	missing	Any	Not permitted	Reject

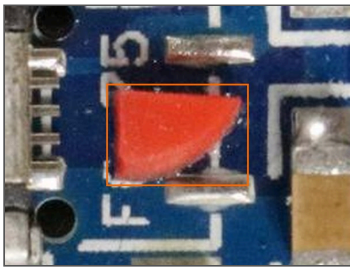


A capacitor absent from its pads (VisA pcb4, anomaly 066).

5.3 Extra component or object

Nothing shall be attached to the part beyond the components of the assembly. A component on pads the design leaves empty, a loose part, a fragment of plastic or a lump of solder or adhesive is nonconforming at any size and in any zone.

Clause	Defect class	Zone	Limit	Disposition
5.3	extra	Any	Not permitted	Reject

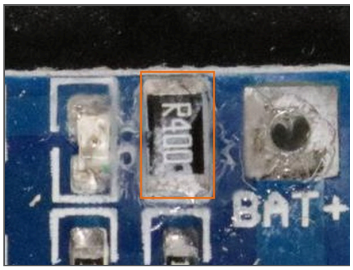


A plastic fragment among the components (VisA pcb4, anomaly 039).

5.4 Misplaced component

Each component shall sit square on its own pads. A component that is shifted off its pads, turned from its drawn orientation or standing on one end is nonconforming at any offset and in any zone.

Clause	Defect class	Zone	Limit	Disposition
5.4	wrong place	Any	Not permitted	Reject

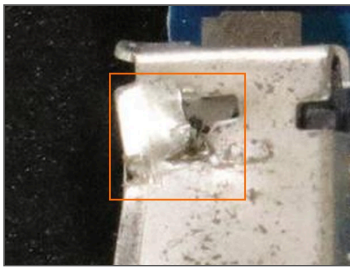


A resistor turned on its pads (VisA pcb4, anomaly 096).

5.5 Damaged component or board

Damage is a break in the material of a component or of the board: a torn or crushed receptacle shell, a cracked or chipped component body, a lifted pad or a gouge through the board surface. It is nonconforming at any size and in any zone.

Clause	Defect class	Zone	Limit	Disposition
5.5	damage	Any	Not permitted	Reject

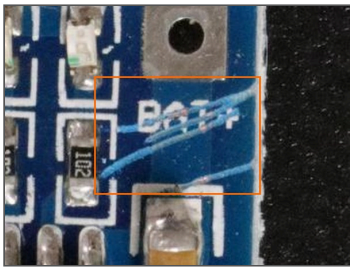


The receptacle shell torn open (VisA pcb4, anomaly 015).

5.6 Scratch

A scratch is a line cut or scored into the surface of the board or of a component. It is nonconforming when its size exceeds the limit of the zone it lies in. Several scratches that touch or cross are measured together as one defect.

Clause	Defect class	Zone	Limit	Disposition
5.6.1	scratch	A	> 1.0 mm	Reject
5.6.2	scratch	B	> 2.0 mm	Reject
5.6.3	scratch	C	> 3.0 mm	Reject

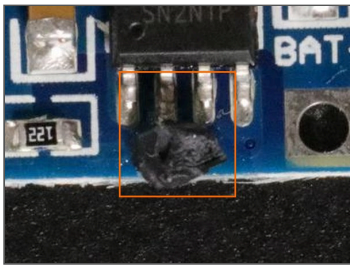


Scratches across a battery pad (VisA pcb4, anomaly 077).

5.7 Burn mark

A burn mark is material charred or blackened by heat, on a component or on the board. It is nonconforming at any size and in any zone.

Clause	Defect class	Zone	Limit	Disposition
5.7	burnt	Any	Not permitted	Reject



A charred lump beside the integrated circuit (VisA pcb4, anomaly 027).

6 Image sources

Every photograph in this document is from the VisA dataset (Zou, Jeong, Pemula, Zhang and Dabeer, "SPot-the-Difference Self-Supervised Pre-training for Anomaly Detection and Segmentation", ECCV 2022), published by Amazon under the Creative Commons Attribution 4.0 licence. Each is cropped, resized or drawn over for this document; the caption of each figure names the category and the image it is taken from, normal or anomaly, by its number.